A Level Mathematics 9709 · P2 Pure Mathematics 2

全套中英文详细学习手册

Detailed bilingual student manual · concept, derivation, worked examples and exam method

本册共 7 个独立课时。每课按三页展开：第一页建立概念、推导方法并完成基础例题；第二页处理 Cambridge 风格应用、英文表达、评分点与易错辨析；第三页扩展考纲边界、变式推演与迁移任务。题目均为依据考纲原创的训练示例，不复制历年试题。

使用范围与版本 | Scope

0580 册依据 Cambridge IGCSE Mathematics 0580（20252027）的 Core/Extended 分层；9709 册依据 Cambridge International AS & A Level Mathematics 9709（20262027）。M2
单独标为拓展/旧体系衔接材料，不作为现行独立考试组件。

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第 1 课 | 代数与模函数 | Algebra and modulus

Lesson 1 of 7 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

|x|是到0的距离，分段线性；方程|f(x)|=a需考虑正负两支。

> English core explanation. Use more than one representation. A formula gives exact relationships, a graph shows shape and intersections, and a table reveals numerical patterns. Factorised, expanded and completed-square forms each expose different information.

为什么方法成立 | Reasoning behind the method

先定位零点分段，或用几何交点；平方前确认不会引入增根。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
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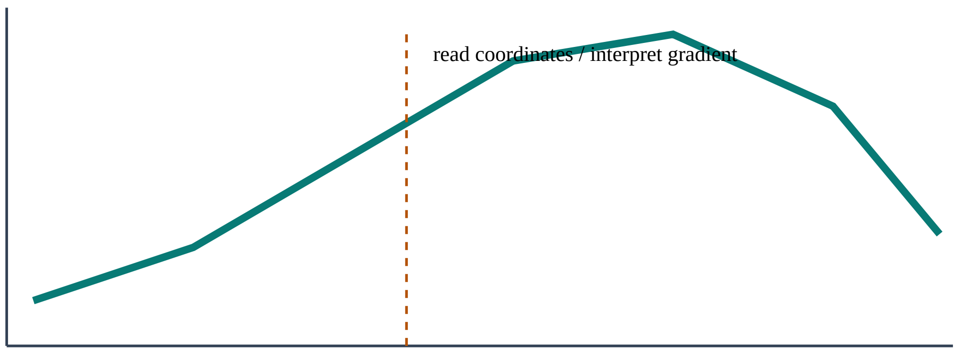

例题一：从题意到完整解答 | Worked example 1

解 |2x3|=7。

2x3=7或7，得x=5或2。

> Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 1 课（续）| 代数与模函数 | Algebra and modulus

例题二：Cambridge 风格应用 | Cambridge-style application

Solve $|x+1|<4$.

$4<x+1<4$, hence $5<x<3$.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得 follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π, e, logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

条件未检查：把只适用于直角三角形、恒加速度、独立事件或正态模型的公式随意套用。修正：公式前写出适用结构。

中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。

只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。

符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。

漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步 | Extension and connection

把本课与相邻知识连接：同一对象通常可从符号、图像、数值表和现实情境四种表示来理解。若代数式得到的符号与图像趋势不一致，应回查运算；若概率或测量答案超出合理范围，应先检查模型而不是盲信计算器。这样的交叉检查是从“会算”走向“会解题”的关键。

> Exam-ready summary. Define or identify → model → calculate with visible steps → interpret → check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

$|x|$是到0的距离，分段线性；方程$|f(x)|=a$需考虑正负两支。 考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。先定位零点分段，或用几何交点；平方前确认不会引入增根。

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变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：解 $|2x3|=7$。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是：$2x3=7$或7，得$x=5$或2。 复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence 推进。上一问的精确形式通常是下一问最简洁的入口。

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“From the given information, …” → “Using …, we obtain …” → “Substituting and simplifying gives …” → “Therefore the required value is …” . For interpretation, write “This means that … in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

Solve $|x+1|<4$.

独立完成后对照：$4<x+1<4$, hence $5<x<3$.

> Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 2 课 | 指数与对数 | Exponentials and logarithms

Lesson 2 of 7 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

对数是指数的逆运算；log laws 只适用于乘积、商和幂。

English core explanation. Algebra records a sequence of equivalent statements. Show the operation that preserves equality, distinguish an expression from an equation, and check restrictions caused by denominators, logarithms, roots or modulus. Substitute solutions back into the original statement.

为什么方法成立 | Reasoning behind the method

先统一底数或两边取对数；解后检查真数大于0。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
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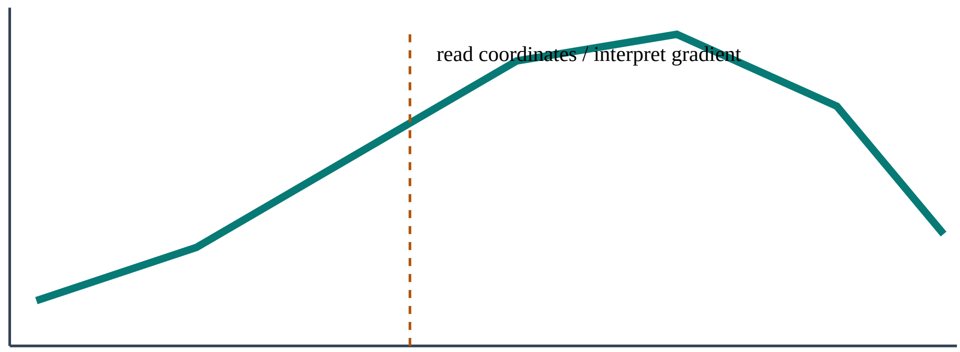

例题一：从题意到完整解答 | Worked example 1

解 3^(2x1)=20。

2x1=ln20/ln3，所以x=[1+ln20/ln3]/2 ≈ 1.863。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 2 课（续）｜指数与对数 | Exponentials and logarithms

例题二：Cambridge 风格应用 | Cambridge-style application

Solve ln(x1)+ln(x+1)=ln8.

ln(x1)=ln8, so x=9. Domain x>1 gives x=3.

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中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。
只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。
符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。
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Detailed English explanation. Algebra records a sequence of equivalent statements. Show the operation that preserves equality, distinguish an expression from an equation, and check restrictions caused by denominators, logarithms, roots or modulus. Substitute solutions back into the original statement.

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独立完成后对照：ln(x1)=ln8, so x=9. Domain x>1 gives x=3.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 3 课 | 三角恒等与方程 | Trigonometric identities

Lesson 3 of 7 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

恒等式对定义域内所有值成立；方程只对某些值成立。

English core explanation. Algebra records a sequence of equivalent statements. Show the operation that preserves equality, distinguish an expression from an equation, and check restrictions caused by denominators, logarithms, roots or modulus. Substitute solutions back into the original statement.

为什么方法成立 | Reasoning behind the method

优先把式子化成同一种函数，使用sinx+cosx=1和tanx=sinx/cosx。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
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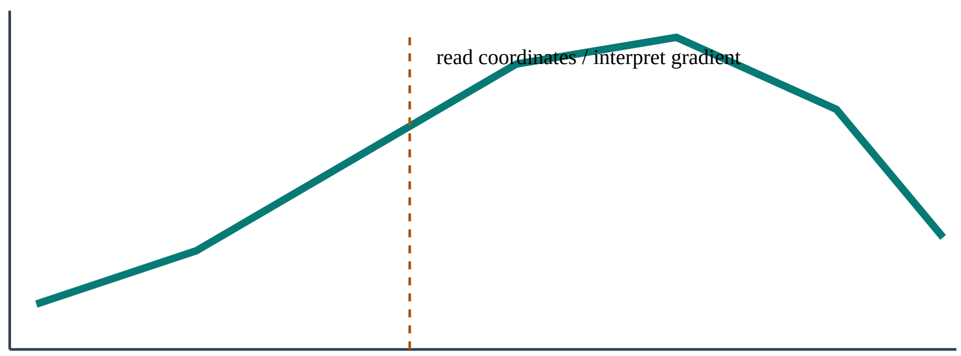

例题一：从题意到完整解答 | Worked example 1

证明（1cosx)/sinx=sinx。

1cosx=sinx，所以左边=sinx/sinx=sinx（sinx≠0处；原式也要求分母非零）。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 3 课（续）｜三角恒等与方程｜Trigonometric identities

例题二：Cambridge 风格应用｜Cambridge-style application

Solve 2cosx3cosx+1=0 for $0 \leqslant x \leqslant 2\pi$.

(2cosx1)(cosx1)=0; $x=\pi/3, 5\pi/3, 0, 2\pi$.

评分点如何产生｜Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得 follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

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只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。
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第三个迁移任务 | Transfer task

Solve 2cosx3cosx+1=0 for 0⩽x⩽2π.

独立完成后对照：(2cosx1)(cosx1)=0; x=π/3,5π/3,0,2π.

> Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 4 课 | 微分法则与应用 | Differentiation applications

Lesson 4 of 7 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

链式、乘积和商法则处理复合表达式；导数模型可给速度、最优化和近似。

English core explanation. Calculus links local change with accumulated change. Identify the inner function or variable, retain the constant of integration when required, and interpret derivatives, stationary points and definite integrals rather than treating them as symbol manipulation.

为什么方法成立 | Reasoning behind the method

先识别外函数与内函数；优化题写目标函数、定义域、驻点与验证。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
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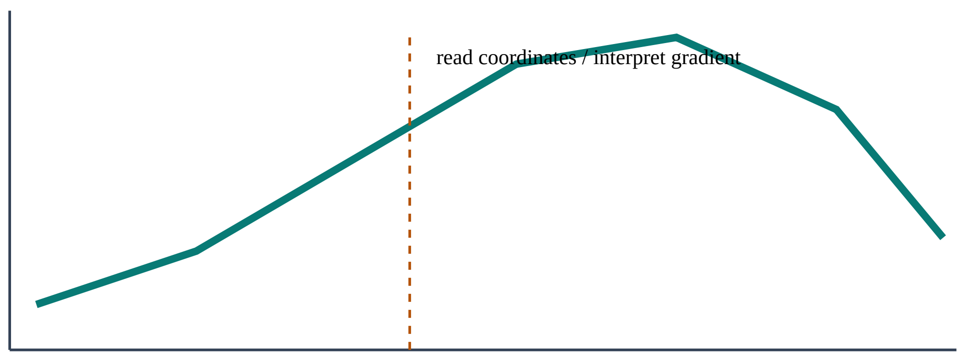


例题一：从题意到完整解答 | Worked example 1

微分 y=(3x1)^5。

dy/dx=5(3x1)^4 × 3=15(3x1)^4。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 4 课（续）｜微分法则与应用｜Differentiation applications

例题二：Cambridge 风格应用｜Cambridge-style application

A rectangle has perimeter 40. Maximise its area.

Let sides x and 20x. A=20xx; A′=202x=0 gives x=10, and A″<0, so max area 100.

评分点如何产生｜Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得 follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting … gives …”, “Therefore …”, “Since the discriminant is zero …”, “The events are independent because …”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π, e, logarithms or radicals by rounded decimals.

易错辨析｜Common errors and repairs

条件未检查：把只适用于直角三角形、恒加速度、独立事件或正态模型的公式随意套用。修正：公式前写出适用结构。

中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。

只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。

符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。

漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步｜Extension and connection

把本课与相邻知识连接：同一对象通常可从符号、图像、数值表和现实情境四种表示来理解。若代数式得到的符号与图像趋势不一致，应回查运算；若概率或测量答案超出合理范围，应先检查模型而不是盲信计算器。这样的交叉检查是从“会算”走向“会解题”的关键。

> Exam-ready summary. Define or identify → model → calculate with visible steps → interpret → check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

链式、乘积和商法则处理复合表达式；导数模型可给速度、最优化和近似。 考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”， 再决定使用哪条规则。先识别外函数与内函数；优化题写目标函数、定义域、驻点与验证。

> Detailed English explanation. Calculus links local change with accumulated change. Identify the inner function or variable, retain the constant of integration when required, and interpret derivatives, stationary points and definite integrals rather than treating them as symbol manipulation.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：微分 $y=(3x1)^5$。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是：$dy/dx=5(3x1)^4 \times 3=15(3x1)^4$。 复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence
推进。上一问的精确形式通常是下一问最简洁的入口。

英文作答骨架 | Useful response frames

“From the given information, …” → “Using …, we obtain …” → “Substituting and simplifying gives …” → “Therefore the required value is …” . For interpretation, write “This means that … in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

A rectangle has perimeter 40. Maximise its area.

独立完成后对照：Let sides x and 20x. A=20xx; A′ =202x=0 gives x=10, and A″ <0, so max area 100.

> Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 5 课 | 积分与面积 | Integration applications

Lesson 5 of 7 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

定积分可求曲线与轴或两曲线间面积；上减下并关注交点。

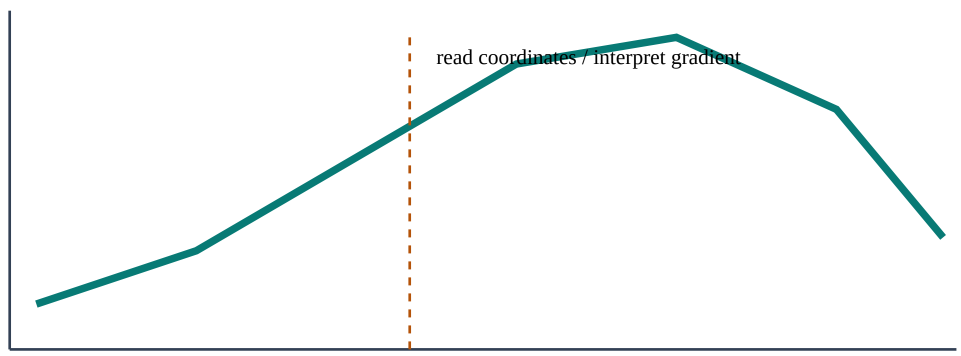
English core explanation. Calculus links local change with accumulated change. Identify the inner function or variable, retain the constant of integration when required, and interpret derivatives, stationary points and definite integrals rather than treating them as symbol manipulation.

为什么方法成立 | Reasoning behind the method

先解交点决定上下关系；含(ax+b)^n可反链式积分。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

求∫(2x+1)^4 dx。

(2x+1)^5/10+C。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 5 课（续）| 积分与面积 | Integration applications

例题二：Cambridge 风格应用 | Cambridge-style application

Find area between y=x and y=x from 0 to 1.

∫(xx)dx=[x/2x/3]=1/6.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得 follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: "Substituting … gives …", "Therefore …", "Since the discriminant is zero …", "The events are independent because …", or "The negative sign shows motion opposite to the chosen positive direction." When a question says hence, use the previous result. When it says exact, do not replace π, e, logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

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中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。
只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。
符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。
漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步 | Extension and connection

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> Exam-ready summary. Define or identify → model → calculate with visible steps → interpret → check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

定积分可求曲线与轴或两曲线间面积；上减下并关注交点。考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。先解交点决定上下关系；含(ax+b)^n可反链式积分。

> Detailed English explanation. Calculus links local change with accumulated change. Identify the inner function or variable, retain the constant of integration when required, and interpret derivatives, stationary points and definite integrals rather than treating them as symbol manipulation.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：求∫(2x+1)^4 dx。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是：(2x+1)^5/10+C。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence
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英文作答骨架 | Useful response frames

“From the given information, …” → “Using …, we obtain …” → “Substituting and simplifying gives …” → “Therefore the required value is …”. For interpretation, write “This means that … in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

Find area between y=x and y=x from 0 to 1.

独立完成后对照：∫(xx)dx=[x/2x/3]=1/6.

> Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 6 课 | 数值解法 | Numerical solution

Lesson 6 of 7 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

迭代法用x_{n+1}=g(x_n)逼近根；收敛并非自动保证。

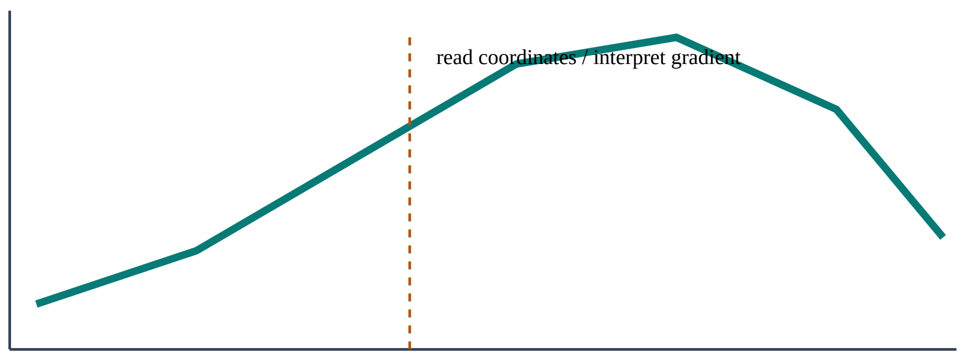
English core explanation. Calculus links local change with accumulated change. Identify the inner function or variable, retain the constant of integration when required, and interpret derivatives, stationary points and definite integrals rather than treating them as symbol manipulation.

为什么方法成立 | Reasoning behind the method

按给定公式逐次计算，报告所需精度；用区间符号变化验证根。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

x_{n+1}=√(2x_n+3)，x=2，求x。

x=√7≈2.6458，x≈2.8795，x≈2.958。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 6 课（续）｜数值解法 | Numerical solution

例题二：Cambridge 风格应用 | Cambridge-style application

Explain how to show a root lies in (a,b).

For continuous f, calculate f(a) and f(b). Opposite signs imply at least one root in the interval.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得 follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting … gives …”，“Therefore …”，“Since the discriminant is zero …”，“The events are independent because …”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π, e, logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

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中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。
只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。
符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。
漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步 | Extension and connection

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Exam-ready summary. Define or identify → model → calculate with visible steps → interpret → check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点丨Scope and connections

迭代法用$x_{n+1}=g(x_n)$逼近根；收敛并非自动保证。 考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。按给定公式逐次计算，报告所需精度；用区间符号变化验证根。

> Detailed English explanation. Calculus links local change with accumulated change. Identify the inner function or variable, retain the constant of integration when required, and interpret derivatives, stationary points and definite integrals rather than treating them as symbol manipulation.

变式推演丨Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核丨Worked reasoning example

重新审视本课基础题：$x_{n+1}=\sqrt{2x_n+3}$，$x=2$，求x。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是：$x=\sqrt{7}\approx 2.6458$，$x\approx 2.8795$，$x\approx 2.958$。
复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装丨How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence
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英文作答骨架丨Useful response frames

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第三个迁移任务丨Transfer task

Explain how to show a root lies in (a,b).

独立完成后对照：For continuous f, calculate f(a) and f(b). Opposite signs imply at least one root in the interval.

> Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 7 课 | P2 综合题 | P2 synthesis

Lesson 7 of 7 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

P2 要求更成熟的代数转换，特别是精确值、定义域和逆向验证。

English core explanation. Translate between words, symbols, diagrams and numerical values. State assumptions, keep exact values during working, and test the final result against the original domain, scale and context.

为什么方法成立 | Reasoning behind the method

每次变形记录依据；含log、root、denominator时做限制检查。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
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4. Keep exact values through intermediate steps; round once at the end.
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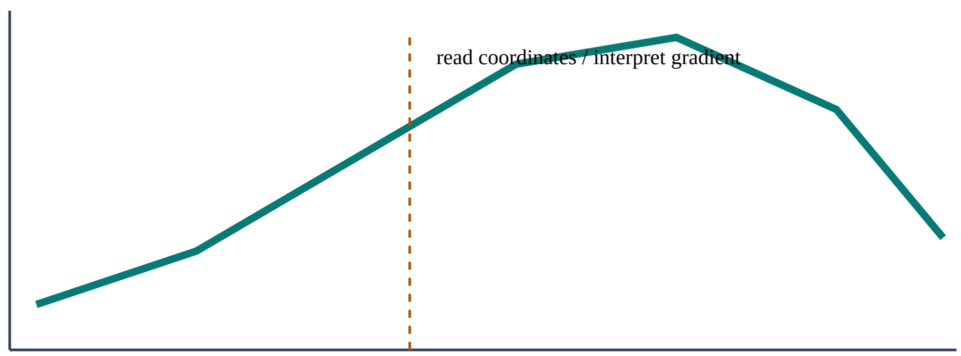

例题一：从题意到完整解答 | Worked example 1

求曲线y=lnx在x=e处的法线。

点(e,1)，切线梯度1/e，法线梯度e；y1=e(xe)。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 7 课（续）| P2 综合题 | P2 synthesis

例题二：Cambridge 风格应用 | Cambridge-style application

Why can squaring create extraneous roots?

Squaring removes sign information, so a value satisfying the squared equation may fail the original equation; substitute back.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得 follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: "Substituting … gives …", "Therefore …", "Since the discriminant is zero …", "The events are independent because …", or "The negative sign shows motion opposite to the chosen positive direction." When a question says hence, use the previous result. When it says exact, do not replace π, e, logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

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中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。

只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。

符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。

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> Exam-ready summary. Define or identify → model → calculate with visible steps → interpret → check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

P2 要求更成熟的代数转换，特别是精确值、定义域和逆向验证。 考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原 “已知量—未知量—约束关系”， 再决定使用哪条规则。每次变形记录依据；含log、root、denominator时做限制检查。

> Detailed English explanation. Translate between words, symbols, diagrams and numerical values. State assumptions, keep exact values during working, and test the final result against the original domain, scale and context.

变式推演 | Variation and transfer

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第三道示范：审题、计算、复核 | Worked reasoning example

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先把文字条件转换成数学关系，再依据本课方法推进。完整结果是：点(e,1)，切线梯度1/e，法线梯度e；y1=e(xe)。 复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence
推进。上一问的精确形式通常是下一问最简洁的入口。

英文作答骨架 | Useful response frames

“From the given information, …” → “Using …, we obtain …” → “Substituting and simplifying gives …” → “Therefore the required value is …” . For interpretation, write “This means that … in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

Why can squaring create extraneous roots?

独立完成后对照：Squaring removes sign information, so a value satisfying the squared equation may fail the original equation; substitute back.

> Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?